

TECHRISE 3.0

Cybersecurity & DevSecOps • Abia Cohort 3.0 • 2026

WEEK 7 • MINI ASSIGNMENT

IDS/IPS • Snort 3.0 • Suricata • Wireshark • Network Security Monitoring

Student Name	_____	Date Submitted	_____
Total Score	_____ / 60 marks	Marked By	_____

Instructions
• Answer ALL questions in ALL sections. Total marks: 60.
• For practical questions — paste your actual terminal output or command. Generic descriptions score zero.
• All lab work must be done on your own authorised lab environment only.
• Section D (Wireshark) is new material — read the introduction carefully before answering.
• No plagiarism — identical answers between students score zero for both.

A	IDS/IPS FUNDAMENTALS — Concepts and Theory	15 marks
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Question 1 (4 marks)

Complete the comparison table below:

Feature	IDS	IPS
What it does when attack detected	_____	_____
Network position	Mirror port (passive)	_____
If the tool fails	_____	Network goes down
Risk level	_____	High — false positive blocks users

Question 2 (3 marks)

Explain the four detection outcome types in security monitoring. Give one real example from your Week 7 labs for each:

True Positive:

False Positive:

False Negative:

True Negative:

Question 3 (4 marks)

Read this Snort 3.0 rule carefully and answer the questions below:

```
alert tcp any any -> $HOME_NET 22 (msg:"SSH Brute Force Detected"; flow:to_server;  
detection_filter:track by_src, count 5, seconds 60; sid:1000002; rev:1;)
```

a. What does 'flow:to_server' do in this rule?

b. What does the detection_filter do — when exactly does this rule fire?

c. Why does this rule use detection_filter instead of firing on every single SSH attempt?

d. What would you change in this rule to make it fire after 3 attempts in 30 seconds instead?

Question 4 (4 marks)

What are the THREE types of IDS/IPS based on detection location? For each one name the type, what it monitors, and which tool from your Week 7 labs represents it:

Type 1:

Type 2:

Type 3:

B	SNORT 3.0 AND SURICATA — Rules and Configuration	20 marks
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Question 5 (6 marks)

Write a Snort 3.0 rule for each scenario below. Each rule must be syntactically correct and on a single line:

- a. Detect ANY ICMP traffic on the network (test rule). Use sid:2000001

- b. Detect anonymous FTP login attempts — match the text 'USER anonymous' in FTP traffic going to your home network on port 21. Use sid:2000002

- c. Detect SQL injection in HTTP URLs — match UNION and SELECT in the HTTP URI using Snort 3.0 sticky buffer syntax. Use sid:2000003

Question 6 (4 marks)

The table below shows a rule written for Snort 3.0. Rewrite it correctly for Suricata and explain what changed and why:

```
# Snort 3.0 version:
alert tcp any any -> any any (msg:"SQL Injection"; flow:to_server,established;
http_uri; content:"UNION", nocase; content:"SELECT", nocase, distance:0;
sid:3000004; rev:1;)
```

Suricata version of the same rule:

Explain ALL differences between the two versions:

Question 7 (4 marks)

Answer these questions about Snort 3.0 and Suricata configuration:

- a. What is the name of the Snort 3.0 configuration file and how is it different from Snort 2.x?

- b. What is the correct Snort 3.0 command to validate your configuration and rules?

- c. What does suricata-update do and why is it important for a production SOC?

- d. What is EVE JSON and why is it more useful than Snort's alert_fast.txt for Splunk integration?

Question 8 (6 marks)

Practical Evidence — paste your actual output for each of the following:

- a) Your five rules from /etc/snort/rules/local.rules (2 marks)

- b) The Snort 3.0 validation output showing rules loaded (2 marks)

- c) One alert from /var/log/snort/alert_fast.txt showing a rule firing (2 marks)

C	NETWORK SECURITY MONITORING — tcpdump and PCAP	10 marks
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Question 9 (4 marks)

Write the exact tcpdump command for each task:

- a. Capture all traffic on interface eth0 and save it to a file called evidence.pcap

- b. Read from a saved PCAP file and show only traffic on port 21

- c. Capture traffic from a specific IP address and display packet contents in ASCII

- d. Show only SSH traffic (port 22) on the network interface in verbose mode

Question 10 (3 marks)

Explain what 'order of volatility' means in digital forensics and list FOUR types of evidence in order from most volatile to least volatile:

Order of volatility means:

Most volatile (collect first):

Second:

Third:

Least volatile (can wait):

Question 11 (3 marks)

You captured a PCAP file during a suspected FTP data theft. Write the command to run Suricata against the PCAP file and explain what advantage this has over only having log files:

Command:

Advantage over log files alone:

D	WIRESHARK BASICS — Network Packet Analysis	15 marks
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Introduction to Wireshark — Read Before Answering	
•	Wireshark is a free, open-source graphical network packet analyser. While tcpdump captures packets in the terminal, Wireshark lets you view, filter, and inspect packets visually in a GUI.
•	Wireshark captures all traffic on a network interface and displays each packet with full protocol breakdown — Ethernet, IP, TCP, HTTP, DNS — all visible layer by layer.
•	Wireshark uses display filters to narrow down traffic. These use a specific syntax different from tcpdump.
•	Wireshark can open PCAP files saved by tcpdump — making it the perfect companion tool for forensic analysis.
•	Install on Kali: <code>sudo apt install wireshark -y</code> Run: <code>sudo wireshark</code>

Question 12 (3 marks)

Wireshark shows packets organised in columns. Match each column to what it displays:

Column Name	Answer (A-E)	Options
No.	_____	A. The protocol used — HTTP, TCP, DNS, FTP
Time	_____	B. A brief description of what the packet contains
Source	_____	C. The sequential packet number
Protocol	_____	D. The IP address the packet came from
Info	_____	E. How many seconds since capture started

Question 13 (4 marks)

Wireshark uses display filters to find specific traffic. Write the correct Wireshark display filter for each task:

a. Show only HTTP traffic

b. Show only traffic from IP address 185.220.101.47

c. Show only FTP traffic on port 21

d. Show only TCP traffic going TO port 22 (SSH)

e. Show only DNS traffic

Hint — Wireshark display filter syntax:

```
http          ← protocol name
ip.addr == 192.168.1.1 ← by IP address
ip.src == 192.168.1.1 ← source IP only
ip.dst == 192.168.1.1 ← destination IP only
tcp.port == 80  ← by port number
tcp.dstport == 22 ← destination port
dns            ← DNS traffic only
ftp           ← FTP traffic only
```

Question 14 (4 marks)

A forensic investigator opens a PCAP file in Wireshark and sees the following packets. Answer the questions below:

No.	Time	Source	Destination	Protocol	Info
1	03:47:15	185.220.101.47	192.168.43.100	TCP	22 → 54321 [SYN]
2	03:47:15	192.168.43.100	185.220.101.47	TCP	54321 → 22 [RST]
3	03:47:16	185.220.101.47	192.168.43.100	TCP	22 → 54322 [SYN]
4	03:47:16	192.168.43.100	185.220.101.47	TCP	54322 → 22 [RST]
5	03:51:44	185.220.101.47	192.168.43.100	FTP	Request: USER anonymous
6	03:51:45	192.168.43.100	185.220.101.47	FTP	Response: 230 Login successful
7	03:51:48	185.220.101.47	192.168.43.100	FTP	Request: RETR /backups/customer_backup.sql.gz
8	03:51:52	192.168.43.100	185.220.101.47	FTP-DATA	FTP Data Transfer 47823912 bytes

a. What is happening in packets 1 and 2? What does [RST] mean?

b. Write the Wireshark display filter you would use to see only the FTP traffic in this capture

c. Packet 8 shows a 47,823,912 byte transfer. Convert this to megabytes and explain what it represents

d. Which Snort 3.0 rule from your Week 7 labs would have detected packet 5? Write the rule message

Question 15 (4 marks)

Answer these short questions about Wireshark:

a. What is the difference between Wireshark and tcpdump? When would you use each one?

b. Wireshark has a 'Follow TCP Stream' feature. What does this show and why is it useful in forensics?

c. A PCAP file was captured with tcpdump using: `tcpdump -i eth0 -w capture.pcap`. How do you open this file in Wireshark?

d. In Wireshark you want to find only HTTP requests containing the word UNION in the URL. Write the display filter:

Hint: `http.request.uri contains "UNION"`

MARKING SCHEME (For Instructor Use)

Question	Marks	Answer Guide
Q1 — IDS vs IPS comparison table	4	IDS: detects/alerts. IPS: blocks. IPS fails = network down. IDS fails = lose detection only. 1 mark per correct cell
Q2 — Four detection outcomes with	3	TP=alert + real attack. FP=alert + no

examples		attack. FN=no alert + real attack. TN=no alert + no attack. 0.5 per definition + 0.25 per lab example
Q3 — Rule analysis questions a-d	4	a) Only inbound SSH. b) Fires after 5 attempts in 60s from same IP. c) Reduces false positives. d) count 3, seconds 30. 1 mark each
Q4 — Three IDS/IPS types	4	NIDS=Snort/Suricata monitors network. HIDS=Wazuh monitors host. NIPS=inline blocking. 1 mark per type + tool + what monitored
Q5 — Three rules written a-c	6	2 marks each. Must be syntactically correct single-line. Rule a: alert icmp. Rule b: content:USER anonymous + nocase. Rule c: http_uri; content:UNION commas for Snort 3.0
Q6 — Snort 3.0 to Suricata conversion	4	Changes: tcp→http, http_uri→http.uri, commas→semicolons. 1 mark per change identified + correct rule written
Q7 — Configuration questions a-d	4	a) snort.lua replaces snort.conf (Lua-based). b) snort -c snort.lua --warn-all. c) Downloads/updates community rules. d) Named JSON fields vs flat text. 1 mark each
Q8 — Practical evidence a-c	6	2 marks each. Must show actual output not descriptions. Rules must have all 5 present. Validation shows rules loaded count. Alert shows PAYLITENG signature name
Q9 — tcpdump commands a-d	4	a) tcpdump -i eth0 -w evidence.pcap. b) tcpdump -r file.pcap port 21. c) tcpdump -i eth0 host IP -A. d) tcpdump -i eth0 -v port 22. 1 mark each
Q10 — Order of volatility	3	Definition: collect in order of what disappears fastest. RAM→network connections→running processes→disk. 1 mark definition + 2 marks order
Q11 — Suricata PCAP replay	3	Command: suricata -c suricata.yaml -r capture.pcap -l output/. Advantage: can replay detection on historical traffic, shareable evidence, no need for live access
Q12 — Wireshark column matching	3	No=C, Time=E, Source=D, Protocol=A, Info=B. 0.5 per correct match — round to nearest whole mark
Q13 — Wireshark display filters a-e	4	a) http. b) ip.addr==185.220.101.47. c) tcp.port==21 or ftp. d) tcp.dstport==22. e) dns. 0.8 marks each, accept reasonable alternatives
Q14 — PCAP analysis a-d	4	a) TCP connection refused — RST=reset, server rejected connection. b) ftp. c) 45.6MB — customer database backup. d) PAYLITENG Anonymous FTP Login. 1 mark each
Q15 — Wireshark short questions a-d	4	a) Wireshark=GUI visual analysis, tcpdump=terminal capture. b) Follow TCP Stream reassembles conversation. c) File→Open. d) http.request.uri contains "UNION". 1 mark each
TOTAL	60	Distinction 51+ • Merit 42+ • Pass 30+ • Refer <30

Snort detects it. Suricata logs it. Wireshark proves it. All three together tell the full story.

All lab work in this assignment was performed on authorised environments only. Nigeria Cybercrimes Act 2015 applies.